AM 463: QUIZ 7
Oct. 31 in class
15 minutes: Closed Book
Answer all problems
Presentation will affect your grade

Question 1:
Consider the model problem for a tracer $B(x, z, t)$ being advected between two walls. Assume the flow is $(U_0, 0)$ (this may not be physical but go with it for this problem).

$$B_t + U_0 B_x = \kappa \left(B_{xx} + B_{zz} \right).$$

If $\kappa \ll 1$ and we make no assumption on the size of U_0, state the basic assumption for the boundary layer scaling and derive the simplified governing equation as we did in the notes. Be sure to define the Reynolds' number-like parameter.

Question 2:

Recall the Ekman layer solution,

$$\begin{aligned} u &= U\left[1 - \exp(-z/\delta)\cos(z/\delta)\right] \\ v &= U\ \exp(-z/\delta)\sin(z/\delta). \end{aligned} \tag{1}$$

where

$$\delta = \sqrt{\frac{2\nu}{f}}.$$

i) Briefly explain what physical situation the Ekman layer describes.
ii) Sketch u and v as functions of z and explain how an Ekman layer differs from the Blasius boundary layer discussed in the slides. You should not need to do any calculations.